XRPL Validator — Daily Engineering Report · 2026-05-05

Period: 2026-05-05 00:00 to 24:00 US Eastern (2026-05-05 04:00 to 2026-05-06 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 4 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-05-05): 0.99841 (34 missed of 21,359). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(validator): refuse to sign until 3 consecutive independent state-hash matches** (`85cb113`, xrpl-validator) VALAUDIT Phase 3 (va-03), simple form. Highest-leverage security fix in the audit. Pre-this-commit, the validator signed network_hash_bytes for every ledger regardless of whether independent state verification had succeeded; the only difference between "VERIFIED" and "network" was a log label. Per the audit, the gate was tautological — a no-op security control.
- **fix(security): enforce 0600 on validator seed; fatal on bad mode or corrupt content** (`032c75a`, xrpl-validator) VALAUDIT Phase 4 (va-04) — seed hygiene. Three behavior changes hardening how the validator's signing seed is loaded/created at startup: 1. Permission check is FATAL on `mode & 0o077 != 0` (group/other-readable) instead of warn-only. Operators ignored the previous warning; the audit's finding was that a key-material file at the wrong mode is a compromise-class issue, not an annoyance.
- **feat(dashboard): surface VALAUDIT Phase 3 gate + new diagnostic counters** (`43f4bd1`, xrpl-validator) The validator has been growing observability for two weeks (silent + mutation divergence detection, RPC-fallback type breakdown, signing gate). The dashboard at halcyon-names.io was still rendering only the metrics from before all of that. This catches it up.

Docs and operations

- **chore(watch_engine): widen warmup threshold for VALIDATIONS SKIPPED color** (`fe0242b`, xrpl-validator)

Independence program

No independence record for UTC day 2026-05-05; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
05-06 01:31	—	xrpl-validator	`85cb113`	feat(validator): refuse to sign until 3 consecutive independent state-hash matches
05-06 01:33	—	xrpl-validator	`fe0242b`	chore(watch_engine): widen warmup threshold for VALIDATIONS SKIPPED color
05-06 01:43	—	xrpl-validator	`032c75a`	fix(security): enforce 0600 on validator seed; fatal on bad mode or corrupt content
05-06 01:59	—	xrpl-validator	`43f4bd1`	feat(dashboard): surface VALAUDIT Phase 3 gate + new diagnostic counters

Times in UTC. A commit appears on the day it was committed to main or merged into it.